UPDATE BRIEF FOR
AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT

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Abstract. The manuscript version of 29 April 2026 (v3) is frozen while under review. This brief records two items for a future revision: one correction that *needs* to be made (a non-load-bearing error in a footnote of Section 2.3), and one addition that *should* be made (the mod-2-quotient nature of two of the three appendix structure-constant tables, recorded *alongside* the tables, which are kept). Section and lemma numbers refer to v3. Every claim below was checked in an independent exact-arithmetic implementation.

1. Required correction: the Section 2.3 footnote

Location. The footnote attached to the sentence "Both $L\bar{s}$ and Ls are sublattices of L" in Section 2.3.
The error. The footnote states two properties of $L\bar{s}$ – that it is a $\mathbb{Z}$-subgroup of L, and that it is "closed under left-multiplication by L" – and then asserts: *"For Ls both properties are immediate from $s \in L$."* The second property is **false for** Ls: a basis-level check shows that 40 of the 64 products $L_i \cdot (L_j s)$ leave Ls, so $L \cdot Ls \not\subseteq Ls$.

The claim is moreover self-contradictory within the paper. Since $Ls \subseteq L$, a closure $L \cdot Ls \subseteq Ls$ would force $Ls \cdot Ls \subseteq L \cdot Ls \subseteq Ls$ – exactly the closure that Remark 4.5 (and the very existence of Table A.3) denies. The non-closure of Ls is the central obstruction the paper resolves; the footnote inadvertently asserts it away.
Why it is not load-bearing. The footnote only needs to justify the sentence it annotates, namely that Ls is a *sublattice* of L. That requires only $Ls \subseteq L$ and that Ls is a subgroup – both genuinely immediate from $s \in L$, since $\ell s \in L \cdot L \subseteq L$. Closure under left-multiplication by L is never used for Ls anywhere in the proof; it was introduced because it is correct and relevant for $L\bar{s}$ (it is Wilson's condition (2)) and then over-extended to Ls by the phrase "both properties." No theorem, lemma, or proposition depends on the false claim.
Suggested fix. Replace "For Ls both properties are immediate from $s \in L$." by, e.g.:

> For Ls, the subgroup property is likewise immediate from $s \in L$. Closure under left-multiplication by L, by contrast, holds for $L\bar{s}$ but *fails* for Ls – and that failure is precisely the obstruction resolved by the twist (Remark 4.5).

This costs one sentence, removes the contradiction, and foreshadows the crux of the proof.

Remark. A related fact, correct and optional: $L\bar{s}$ is in fact a *two-sided* ideal of L ($L \cdot L\bar{s}$, $L\bar{s} \cdot L$, and $L\bar{s} \cdot L\bar{s}$ are all contained in $L\bar{s}$). The paper claims, and needs, only the left-multiplication half; no change is required.

2. The mod-2-quotient nature of Tables A.2 and A.3

Appendix A carries three 64-row tables of integer structure constants: Table A.1 certifies Lemma 4.2 ($L \cdot L \subseteq L$, textbook, cited to Coxeter); Tables A.2 and A.3 certify Lemmas 4.3 and 4.4. **All three**

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tables are kept in the revision – they are the explicit, fully verified certificates. What this section adds, to be placed in Appendix A alongside them, is the structural account of *what* Tables A.2 and A.3 certify: each is, in nature, a statement about one 8-dimensional algebra over $\mathbb{F}_2$.
The mod-2 quotient. The paper already states (Section 2.3, after Wilson) that $L\bar{s}\cap Ls = 2L$; hence $2L \subseteq Ls$ and $2L \subseteq L\bar{s}$, and applying the involution σ (with $\sigma(2L) = 2L$),

$$2L \subseteq \sigma(Ls), \qquad 2L \subseteq \sigma(L\bar{s}).$$

Because $L\cdot L \subseteq L$ (Lemma 4.2) and the octonion product is $\mathbb{Z}$-bilinear, $(a+2c)\cdot b \equiv a\cdot b \pmod{2L}$ for $c \in L$, and likewise in the right factor. The product therefore descends to a well-defined $\mathbb{F}_2$-bilinear product $\bar{\mu}$ on the *mod-2 quotient*

$$\bar{L} := L/2L \cong \mathbb{F}_2^8, \qquad \bar{\mu} : \bar{L} \times \bar{L} \to \bar{L},$$

the octonion algebra over $\mathbb{F}_2$ – equivalently, the E_8 lattice reduced modulo 2. The structure constants of $\bar{\mu}$ are exactly the entries of Table A.1 reduced modulo 2.

Since $2L$ lies in both $\sigma(L\bar{s})$ and $\sigma(Ls)$, those sublattices are *full preimages*, under $L \to \bar{L}$, of $\mathbb{F}_2$-subspaces

$$V := \sigma(L\bar{s})/2L, \qquad W := \sigma(Ls)/2L,$$

and from $\sigma(L\bar{s})+\sigma(Ls) = L$ and $\sigma(L\bar{s})\cap\sigma(Ls) = 2L$ one gets $\bar{L} = V \oplus W$ with $\dim V = \dim W = 4$.
What the two tables certify. Reducing the closure statements modulo $2L$ gives

$$\text{Lemma 4.3} \iff \bar{\mu}(\bar{L}, V) \subseteq V, \qquad \text{Lemma 4.4} \iff \bar{\mu}(W, W) \subseteq W.$$

That is: *Table A.2 certifies that V is a* ***left ideal*** *of the octonion $\mathbb{F}_2$-algebra $L/2L$; Table A.3 certifies that W is a* ***subalgebra*** *of it.* The substantive direction is immediate – for Lemma 4.3, $a \in L$ and $w \in \sigma(L\bar{s})$ give $a \cdot w \in L$ (Lemma 4.2) with class $\bar{\mu}([a], [w]) \in V$, and $\sigma(L\bar{s})$ being the full preimage of V returns $a \cdot w \in \sigma(L\bar{s})$; Lemma 4.4 is identical with W. The descent and both equivalences were checked in exact arithmetic (`python_project/src/verify_mod2_quotient.py`).
A structural by-product. The Conclusion lists as open "a structural reason for Lemma 4.4." The mod-2-quotient reading supplies a partial answer in the requested register: the closure of $\sigma(Ls)$ *is* the statement that the 4-dimensional subspace W is a subalgebra of the 8-dimensional $\mathbb{F}_2$-algebra $L/2L$, and the twist's role is to carry the condition-(3) subspace from $Ls/2L$ (which is not a subalgebra) to W (which is).
Terminology: "mod-2 quotient," not "2-adic." The construction is a *quotient* – the surjection $L \to L/2L$. It should be named the *mod-2 quotient* and kept verbally distinct from a 2-adic *scaling* or *saturation*. A recent and thematically adjacent construction is Corradetti's 2-primary conductor sublattice of E_8 (arXiv:2605.09333, 2026), which *rescales* an order and 2-adically *saturates* a *sub*lattice rather than passing to a quotient. Both works find 2-primary structure governing octonion-derived orders, but the operations differ; the term "mod-2 quotient" keeps them apart.
Remark: a direct framing of Lemma 4.4. By conjugating with σ, Lemma 4.4 is equivalent to the statement that the twisted product closes on Ls, $Ls \cdot_\sigma Ls \subseteq Ls$, even though $Ls \cdot Ls \not\subseteq Ls$ – the octonion-level counterpart of the "two equivalent framings" of Remark 4.6.
Effect on the appendix. Tables A.1–A.3 are *kept* as the explicit certificates. The content of this section is *added* to Appendix A as the structural reading of Tables A.2 and A.3: each certifies an $\mathbb{F}_2$-algebra fact about the mod-2 quotient $L/2L$ – that V is a left ideal, that W is a subalgebra. The integer tables and the mod-2-quotient description are complementary: the tables verify, the quotient explains.

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